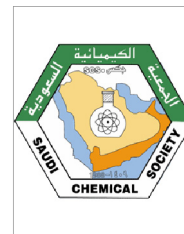




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ORIGINAL ARTICLE

Co₃O₄-Bi₂O₃ heterojunction: An effective photocatalyst for photodegradation of rhodamine B dye



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Abstract Recently, the research on the remediation of aqueous organic pollutants over visible-light-active photocatalysts has got much attention. Therefore, this study reports the fabrication of visible-light-active Co₃O₄-Bi₂O₃ heterojunction photocatalyst for the photodegradation of rhodamine B dye. The Co₃O₄-Bi₂O₃ heterojunction was synthesized by the coprecipitation method and characterized by XRD, EDS, SEM, TEM, TGA, and FTIR. The as-prepared Co₃O₄-Bi₂O₃ heterojunction was utilized as a photocatalyst for the abatement of rhodamine B dye. It was observed that Co₃O₄-Bi₂O₃ showed the best catalytic performance with ~92% degradation of rhodamine B dye than Co₃O₄ and Bi₂O₃ with 14 and 34% removal of rhodamine B dye, respectively. The rate constant for Co₃O₄-Bi₂O₃ catalyzed photodegradation of rhodamine B was 6 times and 3 times higher than the rate constant for Co₃O₄ catalyzed and Bi₂O₃ catalyzed photodegradation of rhodamine B, respectively. The as-prepared Co₃O₄-Bi₂O₃ exhibited the highest catalytic performance at pH 8.

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1. Introduction

The industrial revolution has not only extended the land to mitigate overbuilding but also contributed significantly to environmental pollution (Lo Piccolo and Landi, 2021). The textile industry is one of the industries that pollute the aqueous system due to the discharge of its dyes contaminated wastewater. Several techniques have been pro-

posed for the abatement of wastewater. The removal of organic pollutants from an aqueous system by photocatalysis employing semiconductor metal oxides as photocatalysts play a crucial role in the treatment of wastewater due to its advantages of mild reaction conditions, complete mineralization of pollutants, and low processing cost (S. Li et al., 2017; S. Li et al., 2020b; Chen et al., 2018a; Meng et al., 2019). An ideal photocatalyst for photodegradation of organic pollutants can effectively degrade the pollutants under irradiation of visible light. A narrow band gap semiconductor can be used as a photocatalyst under irradiation of visible light, however, the fast recombination of photo-induced positive holes and electrons inhibits the photocatalytic activity (Li et al., 2019; Huang et al., 2017; Zhong et al., 2017). Therefore, attempts are made to inhibit the recombination of photo-induced positive holes and electrons by developing composite materials. In this respect, several studies have been reported for the synthesis of active visible-light photocatalysts for the treatment of wastewater contaminated with organic pollutants (Li et al., 2018; Gan et al., 2018). The development of visible light-responsive active photocatalyst for dyes contaminated wastewater is a hot topic among the researchers of photocatalysis (Wang et al., 2022; Li et al., 2021; Li et al., 2022a).

The semiconductor bismuth oxide, Bi_2O_3 , has recently attracted the interest of researchers due to its suitable price, stable structure, and suitable band gap energies. The bismuth oxide, Bi_2O_3 , exists in different crystal types: the α - Bi_2O_3 , β - Bi_2O_3 , γ - Bi_2O_3 , δ - Bi_2O_3 , ϵ - Bi_2O_3 and ω - Bi_2O_3 (Zhang et al., 2018; Ho et al., 2013). Among different crystal types, α - Bi_2O_3 , β - Bi_2O_3 have been widely used in catalysis, chemical sensors, fuel cells, photovoltaic cells, and optical thin films. The α - Bi_2O_3 and β - Bi_2O_3 have band gap of 2.85 and 2.58 eV respectively, hence both can be activated under visible irradiation (Chen et al., 2018b; Song et al., 2020). However, the fast recombination photoinduced electron-hole limits the practical application of Bi_2O_3 as a visible light photocatalyst. Therefore, attempts have been made to reduce the rate of recombination of photoinduced electron-hole. The coupling of Bi_2O_3 with semiconductors is one of the effective ways to separate the photoinduced electron and hole. The coupling of Bi_2O_3 with semiconductor metal oxides results in the formation of a heterojunction interface with an electric field between two semiconductors. In this way, the electric field created at the heterojunction assists the transport of charges from one semiconductor to another resulting in an effective separation between the photoinduced charges (Bhaviya Raj et al., 2021; Kaur et al., 2020; Ansari et al., 2021; Balachandran and Swaminathan, 2012). Hence in Bi_2O_3 -semiconductor heterojunctions, advantages such as separation of charges increased lifetime of charges, and enhanced transfer efficiency of the charges to the adsorbed substrate can be achieved. In this study, the development of heterojunction formed by the coupling of Bi_2O_3 with spinel cobalt tetroxide (Co_3O_4) is reported. The spinel cobalt oxide (Co_3O_4) which is composed of Co (II) and Co (III) has been studied extensively due to its extraordinary properties. The high catalytic activity of Co_3O_4 is thought to be due to adsorption of oxygen in different states, oxygen defects, and variance in oxygen holes in Co_3O_4 . Furthermore, the narrow band gap in the range of 1.2–2.1 eV makes Co_3O_4 attractive in photocatalysis for the treatment of wastewater contaminated with organic pollutants (Malefane, 2020; Hu et al., 2019; Luo et al., 2019; Zhao et al., 2019; Rao and Sunandana, 2008). As Bi_2O_3 has oxygen vacancies in its crystals, and Co_3O_4 is rich with oxygen content, therefore Bi_2O_3 promotes mobility and activity of lattice oxygen in Co_3O_4 which results in separation of photoinduced electron-hole pair and ultimately enhances the photocatalytic activity. Hence, we attempted to develop an environmentally friendly and cost-effective method for synthesis of Co_3O_4 - Bi_2O_3 photocatalyst by one step co-precipitation method and evaluate the photocatalytic activity by degradation of rhodamine B dye.

2. Experimental

2.1. Synthesis of photocatalyst

The Co_3O_4 - Bi_2O_3 hetero-structure was prepared by the coprecipitation method. Typically, 12.5 mmol of cobalt nitrate hexahydrate and 12.5 mmol of bismuth nitrate pentahydrate were dissolved in 10 mL (1 M) nitric acid solution under vigorous stirring. Then, 1 M of sodium hydroxide solution was added dropwise to the above-mixed solution under continuous stirring at 60 °C till pH 12 was obtained. The solution was further stirred for 2 h at 60 °C. As a result, a green precipitate was formed. The resultant precipitate was filtered, washed with ethanol, and distilled water till all unreacted ions were eliminated from the prepared precipitate. Then, the washed precipitate was dried at 100 °C overnight. Finally, the dried residue was calcined at 500 °C for 3 h which resulted in black colored Co_3O_4 - Bi_2O_3 hetero-structure.

For comparison, Co_3O_4 and Bi_2O_3 were also prepared. The Co_3O_4 was prepared as follows. A 100 mL solution of cobalt nitrate hexahydrate (0.03 M) was mixed with a 100 mL solution of potassium carbonate (0.06 M) under sitting. The resultant reaction mixture was stirred for 7 h at 70 °C. The obtained residue was filtered, washed, and dried at 100 °C for 12 h. Finally, the dried sample was calcined at 500 °C for 3 h to get Co_3O_4 particles.

The Bi_2O_3 was prepared by adding 1 M sodium hydroxide solution to a solution containing 4.85 g bismuth nitrate pentahydrate in 100 mL till pH 12 was obtained. Concentrated nitric acid was used for the dissolution of bismuth nitrate. The resultant precipitate was filtered, washed, and dried at 100 °C for 12 h. The dried product was calcined at 500 °C to get light yellow Bi_2O_3 particles.

2.2. Characterization

X-rays diffraction, energy dispersive spectroscopy, scanning electron microscopy, thermal gravimetric analysis, and infrared spectroscopy was used for characterization of prepared material using JOEL-JDX-3532 Japan X-ray diffractometer, JEOL-JSM 5910 Japan Scanning electron microscope, JSM5910 UK Energy dispersive X-rays spectrophotometer, Perkin Elmer 6300 TGA analyzer, and Bruker VRTEX70 Infrared spectrophotometer, respectively.

2.3. Catalytic activity

The photocatalytic activities of synthesized particles were determined with photodegradation of rhodamine B dye. Typically, a solution of rhodamine B dye was charged with a predetermined catalyst dose and stirred under sunlight irradiation for 120 min. Progress of photocatalytic degradation was monitored by sampling and analyzing the reaction mixture at a regular time interval. Blank experiments were performed by stirring the dye solution and dye solution with catalyst under irradiation and dark conditions, respectively. UV/Vis spectrophotometer (U-2800, HITACHI, Japan) was used for the measurement of photocatalytic activity.

3. Results and discussion

3.1. Characterization

The formation of Co₃O₄, Bi₂O₃, and Co₃O₄-Bi₂O₃ was confirmed by XRD analysis. The XRD of Co₃O₄ given in Fig. 1 consists of sharp diffraction peaks which show the crystalline nature of Co₃O₄. The diffraction peaks observed at $2\theta \sim 18, 30, 37, 43$, and 58° have been indexed to the spinel structure of Co₃O₄ (Manickam et al., 2016; Rao and Sunandana, 2008; Manigandan et al., 2013). Similarly, the diffraction peaks at $2\theta \sim 12, 26, 33, 36, 41, 46, 54$, and 58° are observed in the XRD of Bi₂O₃. All these diffraction peaks correspond to β -Bi₂O₃. The remaining weak diffraction peaks represent the γ -Bi₂O₃ (Z. Li et al., 2020; Iyyapushpam et al., 2015; Liang et al., 2014; Huang et al., 2016). The existence of characteristic peaks of Co₃O₄ and Bi₂O₃ in the XRD pattern of Co₃O₄-Bi₂O₃ suggest the successful formation of Co₃O₄-Bi₂O₃ heterojunction (Z. Li et al., 2020).

The elemental composition of Co₃O₄, Bi₂O₃, and Co₃O₄-Bi₂O₃ was investigated by energy dispersive spectroscopy using JSM5910, INCA200 UK. Fig. 2 shows the energy-dispersive spectra of the samples. The energy dispersive spectrum of Co₃O₄ given in Fig. 2a shows peaks for Co and O only which confirms the purity of the sample. The EDS analysis showed that prepared cobalt oxide is composed of 87.3 wt% Co and 12.68 wt% O. The energy dispersive spectrum of bismuth oxide (Fig. 2b) shows peaks for Bi, O, and C. The EDS anal-

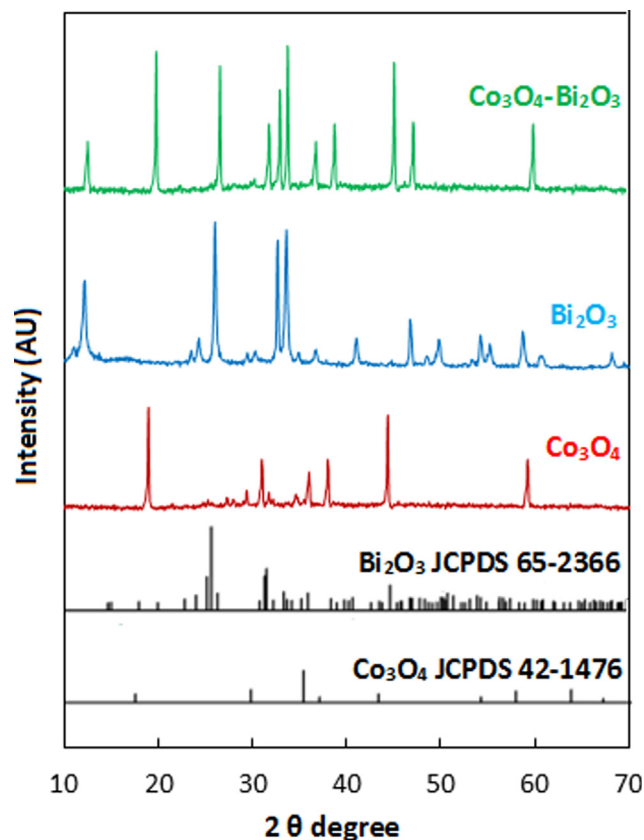


Fig. 1 X-ray diffraction analyses Co₃O₄, Bi₂O₃, and Co₃O₄-Bi₂O₃.

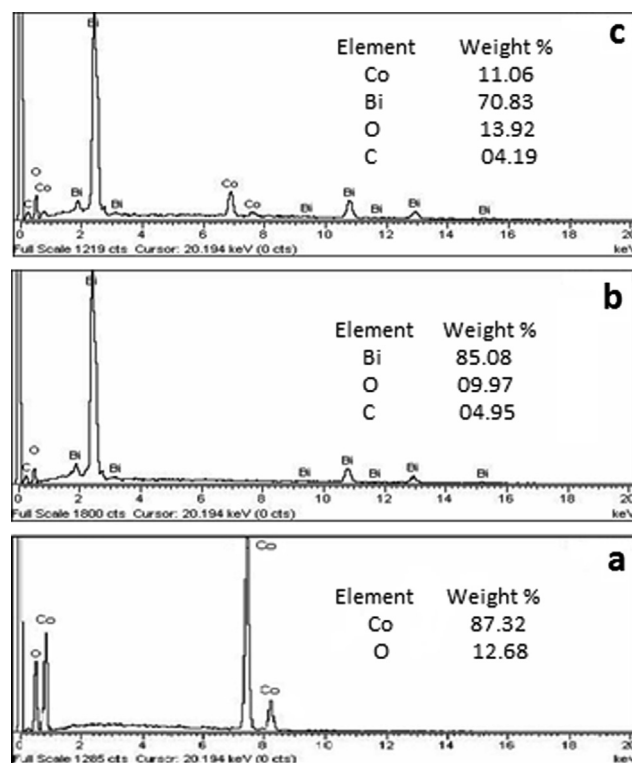


Fig. 2 EDS analysis of Co₃O₄ (a), Bi₂O₃ (b), and Co₃O₄-Bi₂O₃ (c).

ysis showed that prepared bismuth oxide is composed of 58.08, 9.97, and 4.95% Bi, O, and C respectively. The existence of C may be due to impurities in precursor material. Similarly, the energy dispersive spectrum of cobalt oxide-bismuth oxide composite (Fig. 2c) shows that the composite is composed of 11.6, 70.83, 13.92, and 4.19% Co, Bi, O, and C respectively.

The morphology of Co₃O₄, Bi₂O₃, and Bi₂O₃-Co₃O₄ was studied by scanning electron microscopy with JEOL-JSM-5910, Japan scanning electron microscope. JEOL-JSM-420, Japan coating machine was used for mounting and coating the samples with gold foil. The scanning electron micrographs given in Fig. 3a-c show that the particles of as-prepared samples are irregular in shape, non-agglomerated, and dispersed. The non-agglomerated and dispersed particles have enhanced catalytic activity as the active centers are easily accessible to substrate molecules. Fig. 3d and Fig. 3e show the TEM and HR-TEM of Bi₂O₃-Co₃O₄ respectively. Fig. 3d shows that Bi₂O₃-Co₃O₄ is composed of particles with sizes less than 50 nm. The well-defined lattice fringes indicated in HR-TEM show that as prepared Bi₂O₃-Co₃O₄ is highly crystallized (Ding et al., 2012).

Thermal stability of as-prepared samples was estimated by thermal gravimetric analysis using Perkin Elmer 6300 TGA analyzer. As given in Fig. 4, there was only about a 5% loss in weight of the samples up to 700 °C, which is attributed to loss of moisture content. The non-significant weight loss shows the stability of as-prepared samples over a wide range of temperatures.

The typical bonds and functional groups of as-prepared samples were estimated by Fourier transform infrared spectroscopy (FTIR) using Bruker (VRTEX70 series). Fig. 5 shows

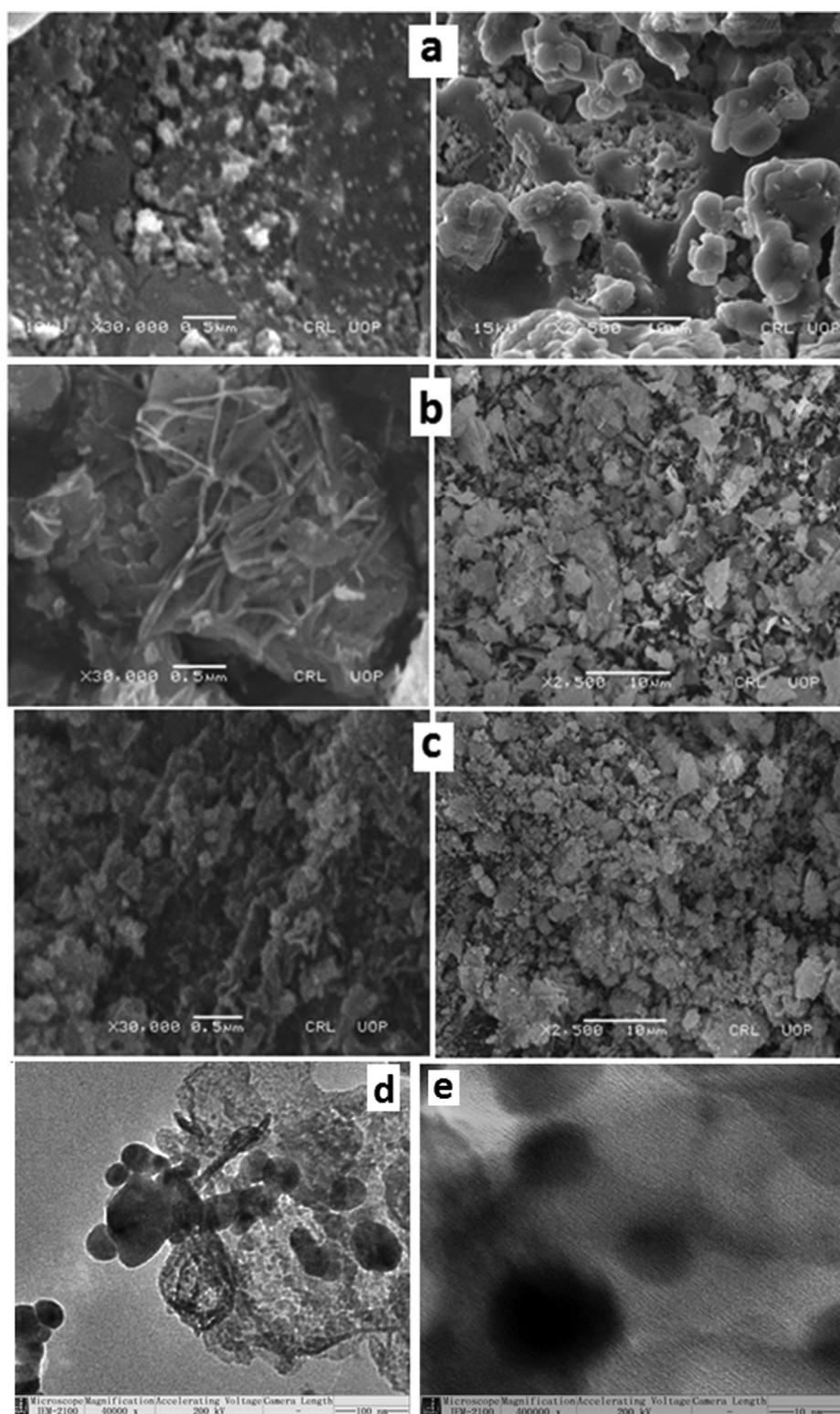


Fig. 3 SEM of Co₃O₄ (a), Bi₂O₃ (b) and Co₃O₄-Bi₂O₃ (c) TEM (d) and HR-TEM (e) images of Co₃O₄-Bi₂O₃.

the FTIR spectra of as-prepared samples in which several peaks can be observed. The absorption peak at 1634 cm^{-1} is attributed to -OH groups present at the surface of Co₃O₄-Bi₂O₃ heterojunction. The absorption bands at $\sim 445\text{ cm}^{-1}$

and 846 cm^{-1} have been assigned to stretching vibrations of Bi-O and Bi-O-Bi bonds, respectively. The absorption bands at ~ 571 and 668 are representative peaks of Co₃O₄ due to Co-O stretching vibrations (Hammad et al., 2016; Ilyas, M,

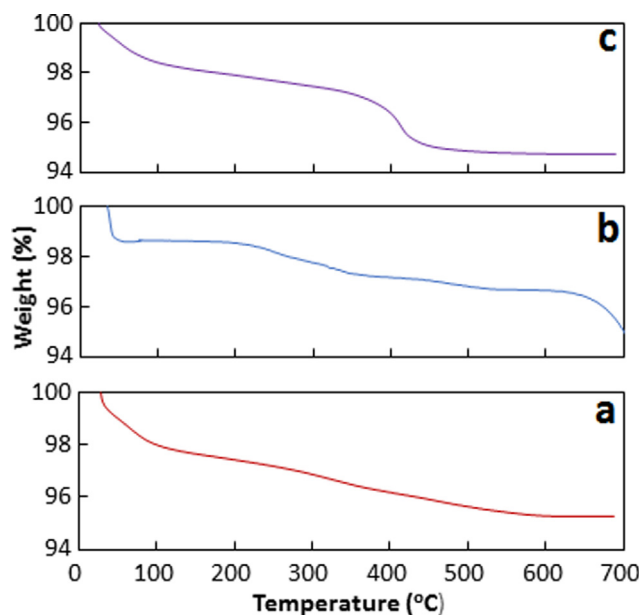


Fig. 4 TGA analysis of Co₃O₄ (a), Bi₂O₃ (b), and Co₃O₄-Bi₂O₃ (c).

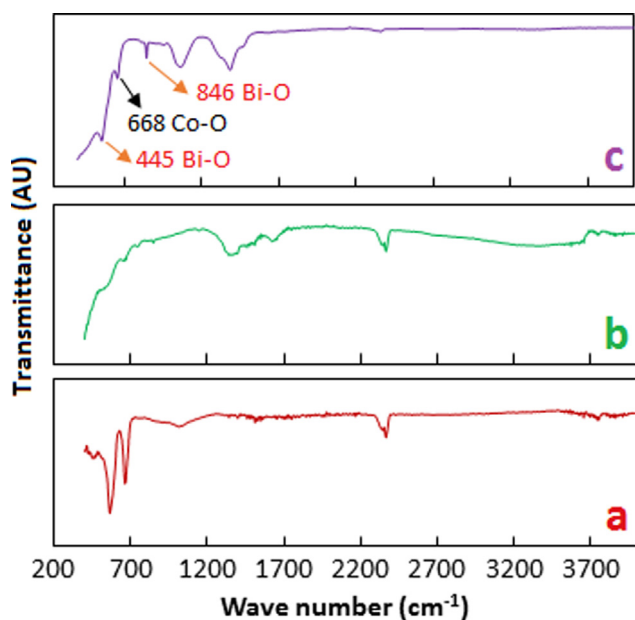


Fig. 5 FTIR analysis of Co₃O₄ (a), Bi₂O₃ (b), and Co₃O₄-Bi₂O₃ (c).

2010; Li et al., 2017a; Saeed et al., 2012; Tang et al., 2018). The FTIR result shows the successful formation of Co₃O₄-Bi₂O₃ heterojunction.

3.2. Catalytic activity

The photocatalytic activity of Co₃O₄, Bi₂O₃, and Co₃O₄-Bi₂O₃ was evaluated by performing photodegradation experiments of rhodamine B dye in the presence of 0.05 g of aforementioned catalysts under visible irradiation. A 50 mL rhodamine B (100 mg/L) was taken in a Pyrex glass beaker and was stirred

under irradiation for 30 min. Then, as a blank reaction, 0.05 g Co₃O₄/Bi₂O₃/Co₃O₄-Bi₂O₃ was added to the dye solution and stirred for 30 min under dark to clarify the effect of adsorption. For evaluation of catalytic performance, the reaction mixture was then irradiated under stirring. The change in concentration of rhodamine B due to catalytic degradation was monitored by measurement of absorbance at wavelength 554 nm using a UV-visible spectrophotometer. Fig. 6a shows the visible spectra of the reaction samples taken from the solution of rhodamine B dye treated with Co₃O₄-Bi₂O₃. As the normalized temporal changes in absorbance (A_t/A_o) of rhodamine B during the photocatalytic process are proportional to normalized concentrations (C_t/C_o) of rhodamine B dye, therefore, the photocatalytic activity was expressed as A_t/A_o vs t as given in Fig. 6b (A_t : absorbance of rhodamine B at time t , A_o : Initial absorbance of rhodamine B, C_t : concentration of rhodamine B at time t , C_o : initial concentration of rhodamine B). It was observed that Co₃O₄-Bi₂O₃ showed the best catalytic performance with ~92% degradation of rhodamine B dye than Co₃O₄ and Bi₂O₃ with 14 and 34% removal of rhodamine B dye, respectively. The obtained results show that coupling of Co₃O₄ with Bi₂O₃ significantly promotes catalytic activity.

A leaching experiment was also performed to confirm whether the photodegradation of rhodamine B is a heterogeneous or homogeneous reaction. For this purpose, 0.1 g Co₃O₄-Bi₂O₃ was suspended in 50 mL distilled water and stirred under irradiation for 120 min. Then, the Co₃O₄-Bi₂O₃ was filtered and a known solution of rhodamine B was added to the

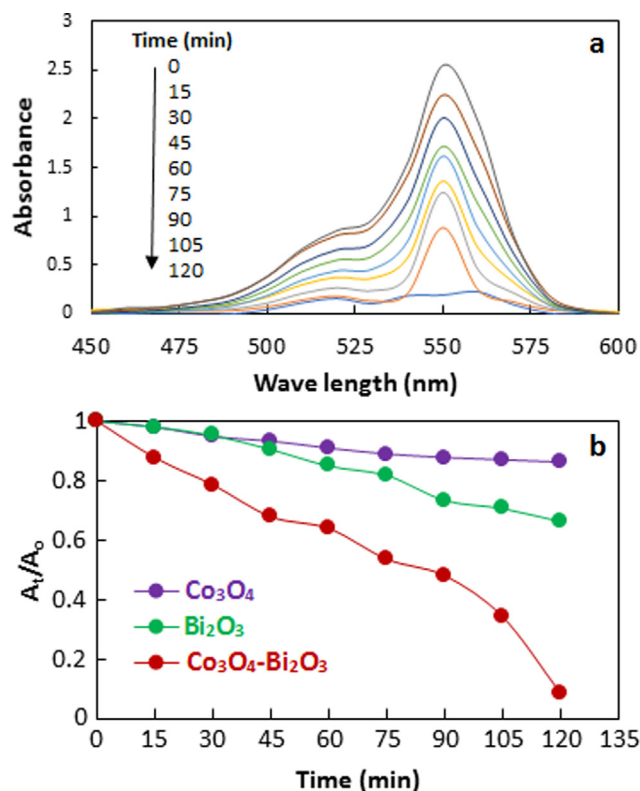


Fig. 6 a) Visible absorption spectra of reaction mixture treated with Co₃O₄-Bi₂O₃ b) Comparison of photocatalytic activity of Co₃O₄, Bi₂O₃, and Co₃O₄-Bi₂O₃ towards photodegradation of rhodamine B dye.

filtrate to get a ~ 100 mg/L dye solution and analyzed with a UV–visible spectrophotometer. Finally, the dye solution was again treated with irradiation and analyzed with a UV–visible spectrophotometer after 120 min. The analysis showed that there was no change in the concentration of the dye. Hence, it is concluded that $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ does not leach to an aqueous medium in this study.

The stability and recycling ability of $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ as photocatalyst was also confirmed. For this purpose, the spent photocatalyst was washed with ethanol and DD water followed by drying. Then, the dried $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ was re-employed for photodegradation of rhodamine B dye under pre-determined experimental conditions. In the same way, the $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ was recycled for three-time. The results showed that there was no significant loss in photocatalytic activity of $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ towards the photodegradation of rhodamine B dye. Hence, it is concluded that the prepared $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ is stable under present experimental conditions and can be recycled again.

$\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ is a second-generation photocatalyst. The second-generation photocatalysts, also called as heterojunctions were developed to overcome the drawback of first-generation photocatalysts (Li et al., 2022b; Liu et al., 2022; Saeed et al., 2022; Anwer et al., 2019). The single-component metal oxides are classified as first-generation photocatalysts. The fast recombination of electron-hole is a basic drawback of first-generation photocatalysts. In second-generation photocatalysts, the photoinduced electrons are confined in the conduction band of one component of heterojunction while the holes are confined in the valence band of the other component. This spatial separation of the photoinduced electrons and holes inhibits their recombination. As a result, active sites are generated at which degradation of organic pollutants takes place. The second-generation photocatalytic materials show light absorbance in the visible region ($\lambda \geq 420$ nm) accompanied with lower band gap energies than first-generation photocatalytic materials. The Mott-Schottky measurement has been reported for band-edge position and conductivity types of Bi_2O_3 and Co_3O_4 having band potentials of 0.28 and 0.5 V vs. Ag/AgCl, respectively. The E_{CB} of Bi_2O_3 and E_{VB} of Co_3O_4 has been estimated as 0.28 and 0.90 V vs NHE, respectively. Accordingly, the E_{VB} of Bi_2O_3 and E_{CB} of Co_3O_4 has been estimated as 2.73 and -1.47 V, respectively (Ma et al., 2022; X. Liu et al., 2021; Xu et al., 2018; Zhu et al., 2018). Hence, the energy band diagram for $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ is constructed as given in Fig. 7. The VB of Co_3O_4 can accept the photoexcited electrons from the CB of Bi_2O_3 due to the short distance between the bands. This transfer of electrons offers a Z-scheme charge transfer model in $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ heterojunction. Hence, the photodegradation of rhodamine B dye over $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ photocatalyst can be described by a direct Z-scheme charge transfer mechanism. According to this mechanism, the photoexcited electrons in CB of Bi_2O_3 flow to the VB of Co_3O_4 under visible light irradiation. The flow of electrons from CB of Bi_2O_3 to the VB of Co_3O_4 reduces the recombination of positive holes and electrons and ultimately it results in enhancement of the photocatalytic activity towards photodegradation of rhodamine B dye (S. Li et al., 2020a; C. Liu et al., 2021; Wang et al., 2020; Zhang et al., 2020).

Hence, the photodegradation of rhodamine B dye in the present study can be described as follow.

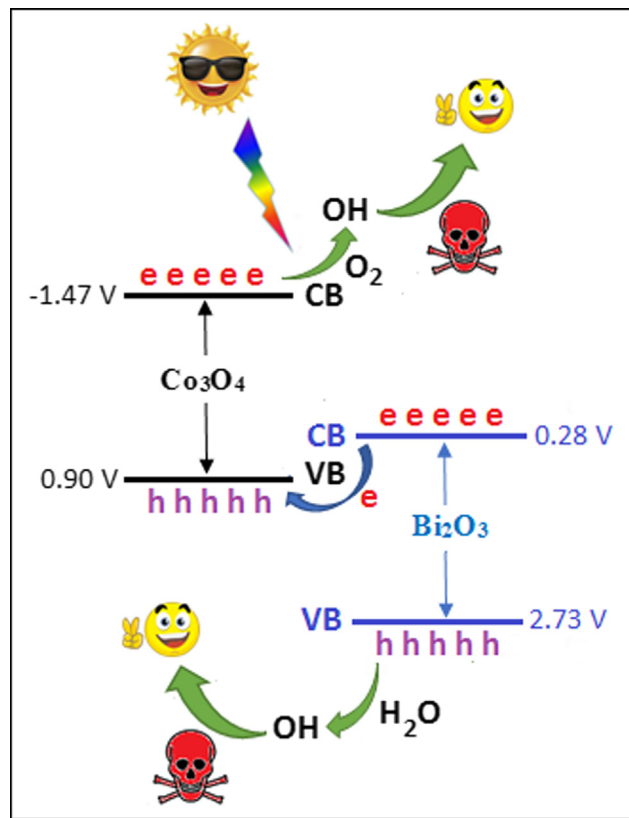
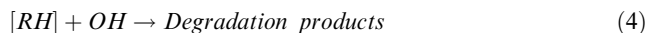
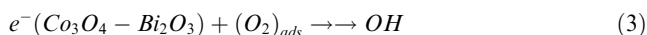
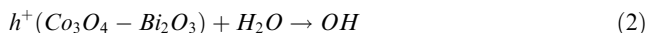
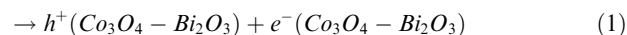
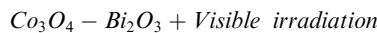


Fig. 7 Schematic diagram of Z-scheme mechanism.



The photoinduced electron, positive hole, and OH radicals are the active species that contribute to the photodegradation of rhodamine B dye. The role played by these species was confirmed by scavenging experiments. For this purpose, EDTA and BQ were separately used as scavengers each of which significantly suppressed the photodegradation activity. Since the EDTA arrests the positive holes, therefore the activity decreased in the presence of EDTA. Similarly, the addition of BQ suppresses photodegradation because it reacts with superoxide anion radicals (Song et al., 2020; Xu et al., 2019).

Based on the above mechanism, the rate expression is written as

$$-\frac{d[\text{RH}]}{dt} = k(\text{O}_2)_{\text{ads}}[\text{RH}]^n \quad (5)$$

$$-\frac{d[\text{RH}]}{dt} = k_{\text{obs}}[\text{RH}]^n \quad (6)$$

$$\ln \frac{A_0}{A_t} = k_1 t (\text{For } n = 1) \quad (7)$$

$$\frac{1}{A_t} = \frac{1}{A_o} + k_2 t \text{ (For } n = 2) \quad (8)$$

k_{obs} , k_1 , k_2 , A_o , and A_t is observed rate constant, 1st order rate constant, 2nd order rate constant, initial absorbance of rhodamine B, and absorbance at time t respectively.

Fig. 8 shows the kinetics treatment of the photodegradation data of rhodamine B. The 1st order rate constant (k_1) and 2nd order rate constant (k_2) are given in Table 1. As the regression coefficient (R^2) values are higher for 1st order kinetics treatment, therefore, it is proposed that the degradation of rhodamine B dye in this study follows 1st order reaction kinetics. The rate constant for Co₃O₄-Bi₂O₃ catalyzed photodegradation of rhodamine B was 6 times and 3 times higher than the rate constant for Co₃O₄ catalyzed and Bi₂O₃ catalyzed photodegradation of rhodamine B respectively. Hence, the formation of Co₃O₄-Bi₂O₃ heterostructure significantly boosts the catalytic performance of Co₃O₄ and Bi₂O₃.

3.3. Effect of pH

The pH of the solution significantly affects the catalytic activity; therefore, the pH was optimized as well in this study. The

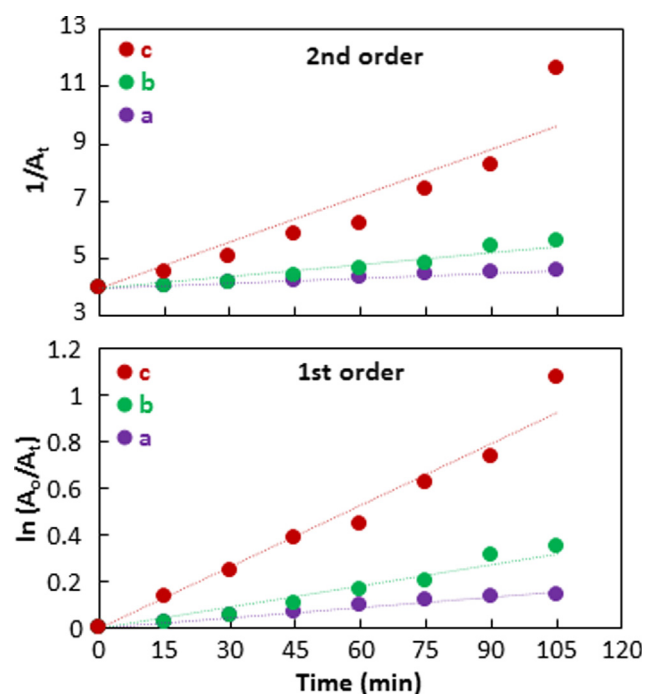


Fig. 8 Kinetics treatment of experimental data with Co₃O₄ (a), Bi₂O₃ (b), and Co₃O₄-Bi₂O₃ (c).

Table 1 The 1st order and 2nd order rate constants for photodegradation of rhodamine B.

Catalyst	$\ln \frac{A_o}{A_t} = k_1 t$		$\frac{1}{A_t} = \frac{1}{A_o} + k_2 t$		*Catalytic efficiency (%)
	k_1	R^2	k_2	R^2	
Co ₃ O ₄	0.0015	0.994	0.006	0.975	14
Bi ₂ O ₃	0.003	0.977	0.0137	0.951	34
Co ₃ O ₄ -Bi ₂ O ₃	0.0089	0.988	0.0541	0.879	92

*Reaction duration: 120 min.

effect of pH on the catalytic activity of Co₃O₄-Bi₂O₃ was evaluated by performing a degradation experiment at pH 4, 6, 8, and 10. Fig. 9 shows the effect of pH on photocatalytic degradation of rhodamine B dye (the numbers given in bar graphs represent the catalytic efficiency). Every experimental cycle was performed with a 0.05 g Co₃O₄-Bi₂O₃ per 50 mL solution (100 mg/L). The reaction duration was 60 min. It was observed that an increase in pH up to 8 favored the catalytic activity. The point of zero charges (PZC) for Co₃O₄-Bi₂O₃ has been reported as pH 8.4 (Ivanova-Kolcheva et al., 2020), hence the surface of Co₃O₄-Bi₂O₃ becomes negative at pH higher than PZC and positive at pH lower than PZC. At pH lower than PZC, both the rhodamine B and surface of the Co₃O₄-Bi₂O₃ are positive, therefore the catalytic activity is low at low pH due to electrostatic repulsion. Similarly, the negative surface of Co₃O₄-Bi₂O₃ at higher pH also opposes the adsorption of rhodamine B, hence the maximum catalytic activity exhibited at pH 8.

3.4. Effect of catalyst dose

The dependence of the photocatalytic activity of Co₃O₄-Bi₂O₃ towards photodegradation of rhodamine B on photocatalyst dosage has been investigated by performing photodegradation experiments with different dosages of Co₃O₄-Bi₂O₃ under identical conditions. In a model experiment, a 50 mL solution of rhodamine B dye (100 mg/L) was treated with a known amount of Co₃O₄-Bi₂O₃ for 60 min. Fig. 10 shows the dependence of catalytic activity on catalyst dosage. It was observed that an increase in catalyst dose from 0.01 to 0.05 g increased the degradation from 18 to 36%, however, further increase in catalyst dose caused a decrease in degradation efficiency. The enhancement in catalytic efficiency with an increase in catalyst dose is due to two reasons: (1) The number of molecules of rhodamine B adsorbed increases with catalyst dosage, (2) the density of Co₃O₄-Bi₂O₃ particles under illumination increases with catalyst dosage. The decrease in catalytic efficiency at higher catalyst dosage is due to the scattering of light. Hence, a catalyst dose of 0.05 g/50 mL of dye solution is found to be the optimum catalyst dosage in this study.

3.5. Effect of concentration

The initial concentration of dye also affects the catalytic activity, therefore the concentration dependence of catalytic activity

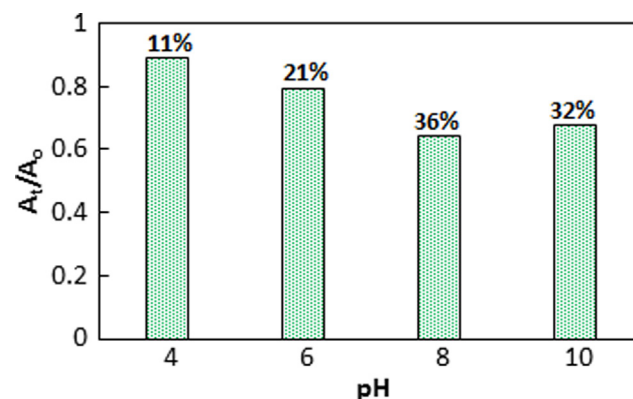


Fig. 9 Effect of pH on the catalytic activity of Co₃O₄-Bi₂O₃ towards photodegradation of rhodamine B.

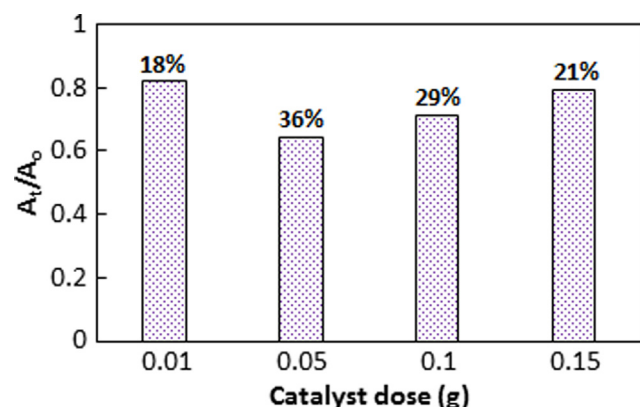


Fig. 10 Effect of catalyst dose on the catalytic activity of $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ towards photodegradation of rhodamine B.

has been also investigated. This investigation was carried out by performing separate photodegradation experiments of rhodamine B dye in the presence of 0.05 g $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ using 100, 200, and 300 mg/L solutions of rhodamine B dye. Fig. 11a shows the effect of concentration on catalytic activity $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ towards photodegradation of rhodamine B dye. It was found that catalytic activity decreased with an increase in the initial concentration of rhodamine B dye. The decrease in catalytic activity with an increase in the concentration of dye is due to three reasons (Adeel et al., 2021; Balcha et al., 2016; Nisar et al., 2022; Saeed et al., 2018):

1. The pathlength of a photon decreases with an increase in the concentration of dye

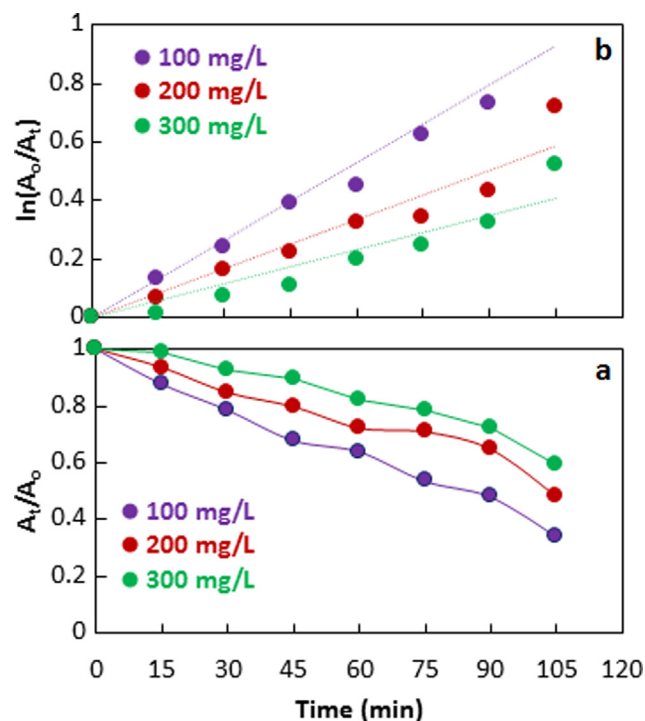


Fig. 11 Effect of concentration of dye on the catalytic activity of $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ towards photodegradation of rhodamine B.

Table 2 Kinetics parameters for $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ catalyzed photodegradation of rhodamine B.

Parameter	100 mg/L	200 mg/L	300 mg/L
k_1 (min^{-1})	0.0089	0.0056	0.0039
R^2	0.988	0.969	0.948
*Catalytic efficiency (%)	92	72	50

*Reaction duration: 120 min.

2. The rhodamine B dye absorbs photons significantly than catalyst at a higher concentration of dye
3. The ratio of OH radicals to rhodamine B molecules decreases with an increase in concentration

The degradation data given in Fig. 11a was analyzed for kinetics studies using kinetics equation (7). Fig. 11b shows the fitting of the 1st order kinetics model (equation (7)) to experimental data. The kinetics parameters given in Table 2 shows that the rate constant decreases with an increase in the initial concentration of rhodamine B dye.

4. Conclusions

Herein, we prepared the $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ heterojunction as visible-light responsive photocatalyst by coprecipitation method for photodegradation of rhodamine B dye successfully. The $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ heterojunction was characterized by XRD, EDS, SEM, TGA, and FTIR. The as-prepared $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ heterojunction was utilized as a photocatalyst for photodegradation of rhodamine B dye using a 100 mg/L solution. The photocatalytic activity of $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$, Co_3O_4 , and Bi_2O_3 towards photodegradation of rhodamine B dye was found as 92, 14, and 34%, respectively. The 1st order and 2nd order kinetics models were applied to the data of photocatalytic degradation of rhodamine b dye. The effect of pH, catalyst dose, and initial concentration of rhodamine B dye on photocatalytic performance was investigated. The $\text{Co}_3\text{O}_4\text{-Bi}_2\text{O}_3$ heterojunction was found as an efficient visible-light-driven photocatalyst for photodegradation of rhodamine B dye.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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